**Project Presentation of Research Paper**

# Mental Health Assessment Using Sentiment Analysis

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# Mental Health Assessment using Sentiment Analysis

Mental Health Assessment using Sentiment Analysis is a sophisticated NLP-driven research project designed to automate the detection of early psychological distress in digital communications. Developed using Python, Scikit-learn, and advanced text-processing libraries, the system provides a non-invasive way to analyze social media "digital footprints" and identify critical emotional trends, helping bridge the gap between digital expression and clinical intervention.

In addition to core sentiment analysis, the framework integrates a sophisticated multi-class classification module capable of predicting seven specific mental states, including Depression, Anxiety, and Suicidal Ideation. By leveraging TF-IDF vectorization and a mathematically optimized Linear SVM model, the platform efficiently processes large-scale datasets—exceeding 52,000 records—to deliver accurate, clinically relevant insights while filtering out linguistic noise.

By combining automation with high-precision machine learning, this research reduces the barriers to mental health screening and supports proactive intervention. Its scalable architecture is designed for seamless integration into healthcare monitoring systems, allowing for future enhancements like temporal tracking to monitor a user’s well-being over time and ensure timely support for those in need.

# Problem Statement

**Challenges in Traditional Mental Health Screening**

- Clinical mental health support is often inaccessible or heavily stigmatized, preventing many individuals from seeking timely help.
- Traditional diagnosis relies on self-reporting and manual assessments during infrequent visits, which often miss the "digital footprints" of distress left online.
- Most existing automated models are limited to simple binary classification (e.g., "Depressed" vs. "Normal"), failing to capture the complexity of distinct psychological disorders.

**Solution –**

- An automated, non-invasive NLP framework designed to detect early signs of psychological distress from social media and digital text.
- Provides a comprehensive multi-class classification system capable of predicting 7 specific mental states, including Depression, Anxiety, and Suicidal Ideation.
- Utilizes advanced preprocessing and TF-IDF vectorization to identify rare, emotionally significant vocabulary that traditional methods might overlook.
- Employs a mathematically optimized Linear SVM model to achieve high accuracy (76.42%) and draw clear boundaries in complex linguistic data.
- Serves as a scalable screening tool for clinicians, enabling proactive intervention and data-driven monitoring of a patient's mental well-being.

# Proposed System Framework

**Mental Health Assessment System** is a Python-based NLP framework that provides an end-to-end solution for automated psychological distress detection. It is designed to handle large-scale textual data, enabling seamless text cleaning, feature extraction, and high-precision classification of multiple mental health states.

**Data Processing Modules**

- Securely input textual data or social media "digital footprints.“
- Apply advanced noise removal (URLs, symbols, and case normalization).
- Perform Lemmatization to reduce words to their linguistic roots.
- Filter out "stop-words" to focus on high-impact emotional vocabulary.

**Analysis & Classification Modules**

- Execute feature engineering using TF-IDF vectorization.
- Process data through an optimized Linear SVM machine learning pipeline.
- Classify text into 7 specific categories (Normal, Depression, Anxiety, etc.).
- Generate performance metrics and "Confusion Matrices" for clinical validation.

# Technologies Used

**Programming &Environment:**

- **Python**: Primary language used for its extensive support in AI and Machine Learning.
- **Jupyter Notebook / VS Code**: Interactive environments for model development and debugging.

**Machine Learning & NLP Libraries :**

- **Scikit-learn**: Core library used for implementing Naive Bayes, Random Forest, and Linear SVM.
- **NLTK / SpaCy**: Utilized for advanced Natural Language Processing tasks like Lemmatization and stop-word removal.

**Data Handling & Engineering :**

- **Pandas & NumPy**: Essential for managing the large-scale dataset of 52,681 records and performing matrix operations.
- **TF-IDF Vectorizer**: Used to transform raw text into a weighted mathematical format for model training.

**Documentation & Research Tools:**

- **Overleaf (LaTeX):** Used for professional research paper formatting and mathematical equation rendering.
- **GitHub:** For version control and managing the research project repository.

# System Architecture

**Data Acquisition Layer (Input)**

**Dataset** → High-volume corpus of 52,681 records.

**Input** → Social media "digital footprints" and textual behavioral markers.

**NLP Computational Layer (Processing-Python)**

**Preprocessing** → Lemmatization, noise removal, and stop-word filtering.

**Feature Engineering** → TF-IDF vectorization (Capped at 5,000 features).

**Predictive Intelligence Layer (Output-Linear SVM)**

**Classification**→ Multi-class categorization into 7 mental health states.

**Validation** → Evaluation of precision, recall, and F1-scores.

# Data Flow & Feature Extraction

**Data Intake & Aggregation:**

- Collection of social media "digital footprints" and behavioral text markers.
- Utilizing a high-volume dataset of 52,681 records for broad clinical coverage.

**Systematic Preprocessing:**

- **Noise Removal:** Eliminating URLs, special characters, and non-essential symbols.
- **Normalization:** Implementing Lemmatization to reduce words to their linguistic roots.
- **Filtering:** Removing stop-words to isolate high-impact emotional vocabulary.

**Feature Engineering (TF-IDF):**

- **Vectorization:** Converting raw text into a weighted numerical matrix.
- **Statistical Weighting:** Applying TF-IDF to prioritize rare, clinically significant terms (e.g., "hopeless," "insomnia").
- **Dimensionality Control:** Capping features at the top 5,000 terms to ensure high-speed computational efficiency for the machine learning models.

# Experimental Results & Metrics

**Accuracy Benchmarking**
Comparison of three distinct models:
Multinomial Naïve Bayes (65.35%)
Random Forest(71.20%)
Linear SVM (76.42).

**Results & Metrics**

**Error Analysis & Refinement**
Identifying vocabulary overlaps between “Stress” and “Depression” to improve future clinical sensitivity.

**Multi-Class Precision**
Achieved a high precision of 0.83 for specific clinical categories like bipolar Disorder and Anxiety.

# Advantages of the Research

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**Non-Invasive Screening :** Early detection using existing social media text.

**Mathematically Robust:** Optimized Linear SVM ensures high prediction accuracy.

**Clinical Scalable :** Handles massive datasets with low computational overhead.

**Objective Markers:** Data-driven metrics eliminate subjective diagnostic bias.

# Conclusion

Successfully validated a 7-class sentiment analysis framework for mental health screening.

Effectively processed a large-scale dataset of 52,681 records using advanced NLP and TF-IDF engineering.

Proved Linear SVM as the superior architecture with a validated accuracy of 76.42%.

Developed a scalable, objective tool that bridges the gap between digital expression and proactive clinical intervention.

# Thank You